

Polar Forms of Conic Sections

Date_____ Period____

Each polar equation describes a conic section with a focus at the origin. Find the eccentricity, find the equation of the directrix associated with the focus at the origin, and classify the conic section.

1) $r = \frac{0.8}{1 - 0.8\sin \theta}$

2) $r = \frac{1}{1 - \cos \theta}$

3) $r = \frac{2}{1 + 2\cos \theta}$

4) $r = \frac{10}{2 - 2\sin \theta}$

5) $r = \frac{8}{4 - 1.6\sin \theta}$

6) $r = \frac{14.4}{2 - 4.8\cos \theta}$

Each problem describes a conic section with a focus at the origin. Classify the conic section, and write the polar equation in standard form.

7) Eccentricity: 1
Directrix for focus at origin: $y = -6$

8) Eccentricity: 1.5
Directrix for focus at origin: $x = 2$

9) Eccentricity: 2.1
Directrix for focus at origin: $y = -3$

10) Eccentricity: 0.5
Directrix for focus at origin: $y = -2$